

EXPERIENCE LAB INTERNSHIP PROGRAMME

Week 5: Project Summary

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HealthConnect Clinic Experience Lab*Data Science Track — Data Preparation, Feature Engineering & Baseline Model*

Programme	AnalystLab Africa Experience Lab
Week	Week 5
Project	HealthConnect Clinic Experience Lab
Track	Data Science

1. What Was Planned for Week 5

The plan was to move from the Week 4 problem definition into practical implementation: preparing the modelling dataset, exploring feature relationships with visual evidence, engineering new features, establishing a sound train/test strategy, and building and evaluating a baseline classification model.

2. What Was Actually Completed

- Reviewed and re-confirmed the Week 4 problem definition, target variable, and cancellation-handling decision.
- Completed a full data preparation pass: checked data types, missing values, duplicates, and categorical consistency; removed irrelevant variables; imputed missing values; encoded categorical features.
- Produced four decision-linked visualizations (target balance, prior no-shows, reminder effect, booking lead time), each tied to a specific preprocessing or modelling decision.
- Engineered three new features: `prior_no_show_rate`, `booking_lead_category`, and `has_prior_history`.
- Implemented a patient-aware train/test split using `GroupShuffleSplit`, verified to have zero patient overlap between sets.
- Built and evaluated a baseline Logistic Regression model.

3. Key Findings or Development Outcomes

- Booking lead time showed the strongest relationship with no-show behavior: no-show rate rises from 29.5% (short lead time) to 63.9% (long lead time).
- Reminders are associated with a modest reduction in no-show rate (49.9% with a reminder vs 54.6% without).
- The baseline model achieved 63.1% accuracy, 66% recall, and 62% precision for the No-Show class, and a ROC-AUC of 0.677, a meaningful improvement over random guessing.

4. Major Challenges Encountered

- The originally flagged data leakage risk for `waiting_time_minutes` did not hold up empirically; the variable turned out to be non-informative rather than leaking future information, requiring a documented correction to the Week 4 reasoning.
- Ensuring a genuinely patient-aware split required using `GroupShuffleSplit` rather than a standard `train_test_split`, since many patients appear multiple times in the dataset.

5. Important Decisions Made and Why

- Retained the Week 4 target variable and cancellation-handling approach unchanged, since Week 5 analysis did not surface evidence to revise it.
- Corrected the reasoning behind excluding `waiting_time_minutes` from data leakage to a non-informative feature, based on empirical evidence rather than assumption.
- Chose Logistic Regression as the baseline model specifically for its interpretability, prioritizing a transparent first benchmark over raw performance.

6. Changes to the Week 4 Approach

The only substantive change was the reclassification of `waiting_time_minutes`'s exclusion reason, from a data leakage risk to a non-informative feature, based on the empirical group comparison performed in Week 5. The practical outcome (exclusion) is unchanged; only the justification was corrected.

7. Cross-Track Collaboration Completed

Identified a conceptual alignment point with the Data Analytics track: both tracks are examining the same underlying variables (reminders, booking lead time) for different purposes, descriptive KPIs versus predictive features, and the feature engineering priorities in this notebook reflect that shared focus.

8. Remaining Work

- No hyperparameter tuning has been performed on the baseline model.
- No alternative model types (Random Forest, Gradient Boosting) have been tested yet.
- Feature importance has not yet been formally extracted from the trained model.

9. Proposed Focus for Week 6

Week 6 will focus on testing tree-based models against the Logistic Regression baseline, performing hyperparameter tuning on the best-performing model, examining feature importance to confirm the strongest predictors, and considering classification threshold adjustments based on HealthConnect's operational priorities.